

Fundamentals of Solid State Physics

Magnetic Properties

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Last Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- Electronic Properties (Week 4–12)
- Thermal Properties (Week 13)
- **Optical Properties (Week 14)**
 - ▣ **Origin of Dielectric constant (ϵ) and Refractive index (n)**
 - ▣ **Optical absorption, reflection, refraction, emission**
- Magnetic Properties (Week 15)

This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- Electronic Properties (Week 4–12)
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- **Magnetic Properties (Week 15)**
 - **Origin of Magnetics**
 - **Diamagnetism, Paramagnetism, Ferromagnetism**
 - **Superconductivity**

References

- <https://cse.umn.edu/irm/1-definitions-and-units>
- <https://www.britannica.com/science/magnetism/Magnetic-properties-of-matter>

History of Magnetism



lodestone
磁铁矿
Greece
600 B.C.



compass
司南
China
1100 A.D.

Applications of Magnetism



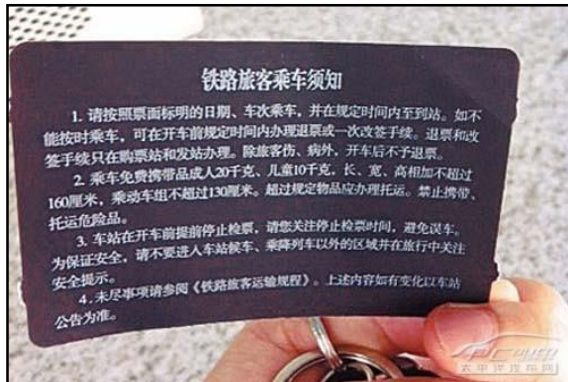
Compass



Hard Drive



Wind Turbine



ID ticket



MagLev 磁悬浮



MRI

Nobel Prizes in Magnetism

- 1902 Zeeman effect
- 1943 Magnetic moment of proton
- 1944 Magnetics of atomic nuclei
- 1952 Nuclear magnetic resonance (NMR)
- 1955 Magnetic moment of electron
- 1970 anti-ferromagnetism and ferri-magnetism
- 1972 BCS theory of superconductivity
- 2007 Giant magnetoresistance
- ...

incomplete list ...

Outline

- Maxwell's Equations

- H, B, M, μ_r
- Magnetic Susceptibility 磁化率 χ

- Origin of magnetism

- spin of electrons, orbital angular momentum, external field
- nuclear magnetic momentum

- Types of magnetism

- **D**iamagnetism 抗磁性
- **P**aramagnetism 顺磁性
- **F**erromagnetism 铁磁性
- **A**ntiferromagnetism 反铁磁性
- **F**erri magnetism 亚铁磁性

Electrodynamics

- Maxwell's Equations

$$\nabla \cdot \mathbf{D} = \rho_V$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

Electrodynamics

■ Maxwell's Equations

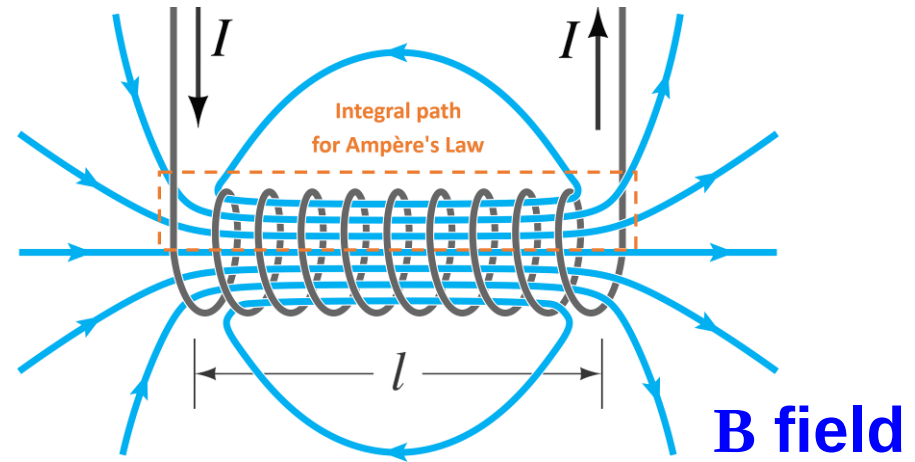
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$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

Solenoid (螺线管)



$$\mathbf{B} = \mu_0 \mu_r n I$$

Electromagnet: Magnetic field is produced by electric currents. (Ampere's law)

Electrodynamics

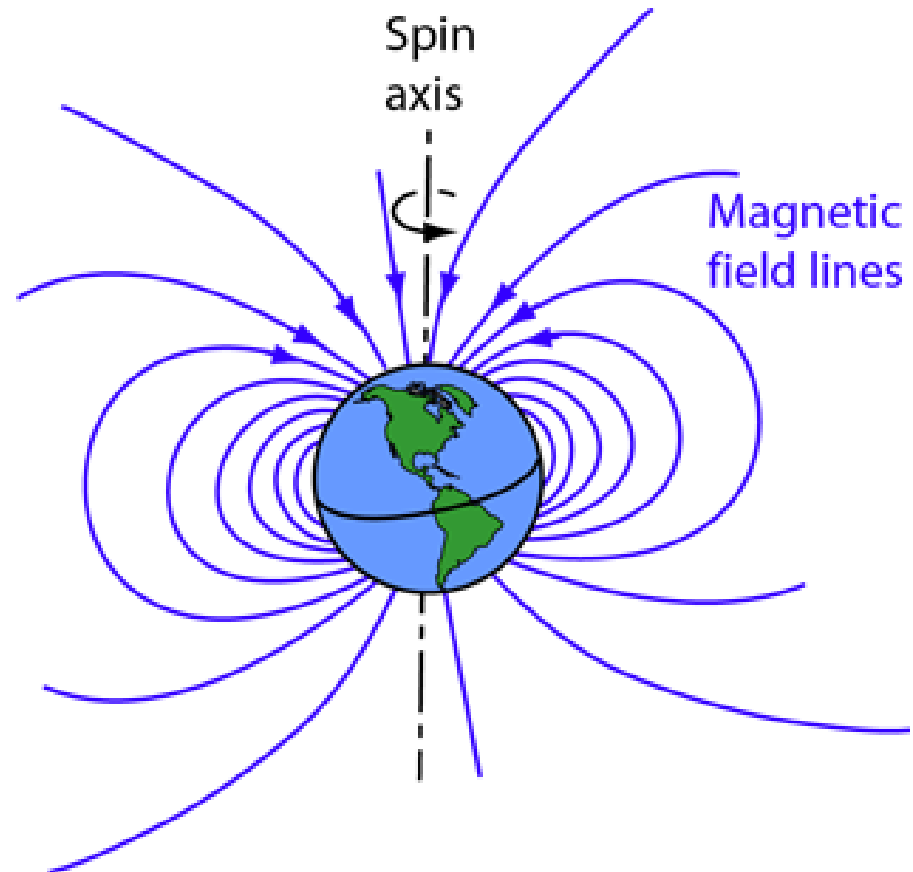
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$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$



Our earth is a big electromagnet

Electrodynamics

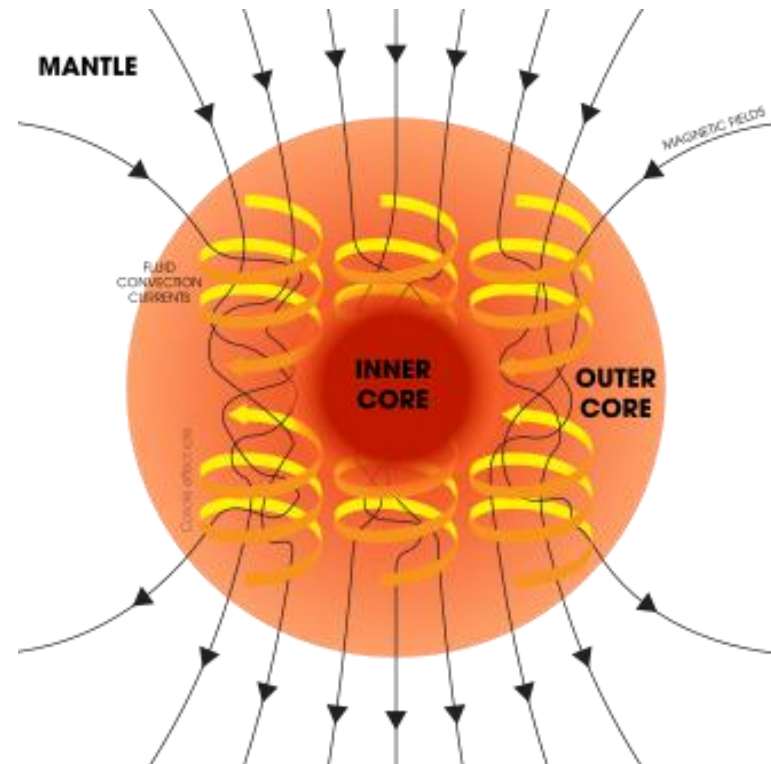
■ Maxwell's Equations

$$\nabla \cdot \mathbf{D} = \rho_V$$

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$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$



The dynamo mechanism

Electrodynamics

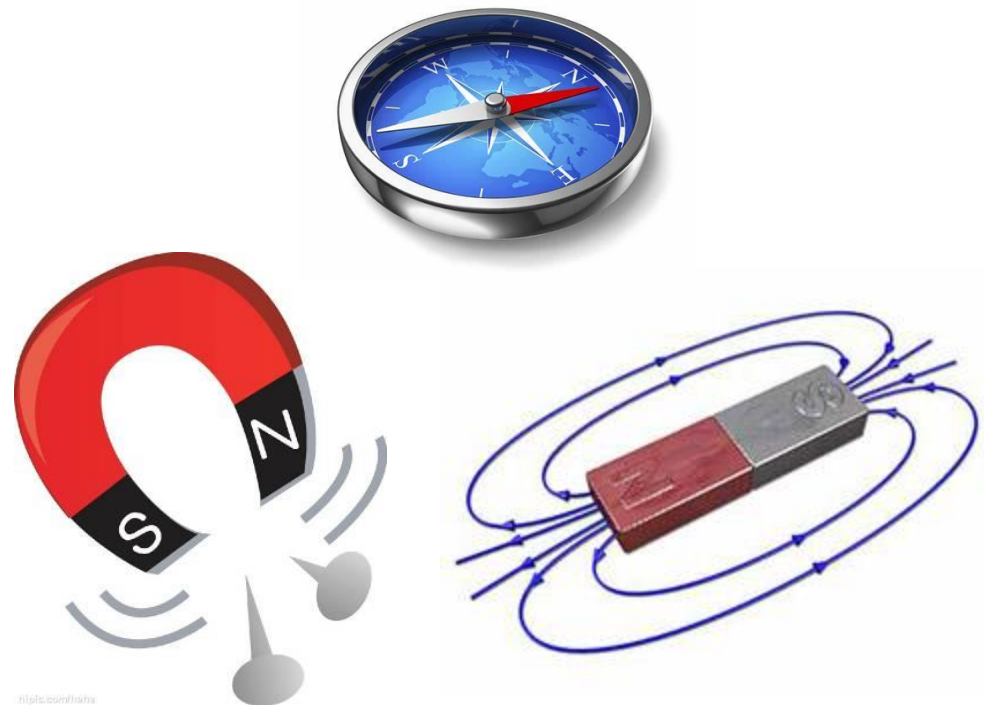
■ Maxwell's Equations

$$\nabla \cdot \mathbf{D} = \rho_V$$

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$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

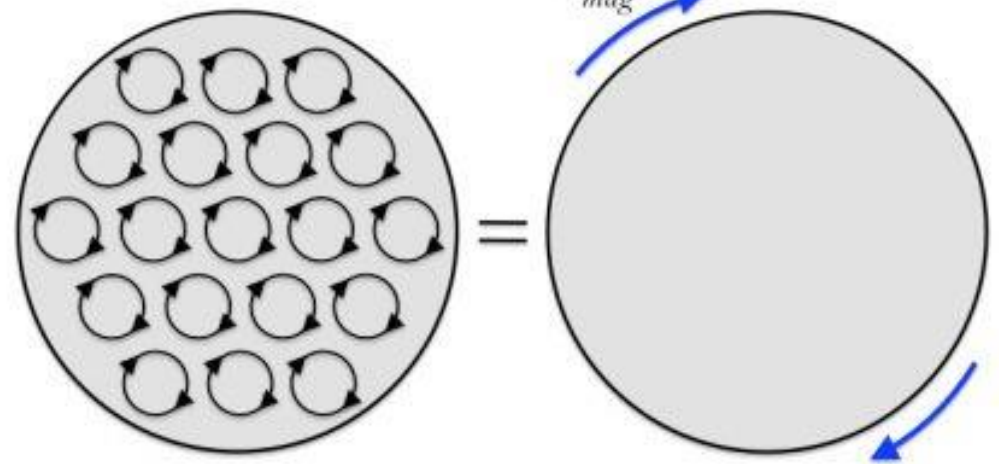
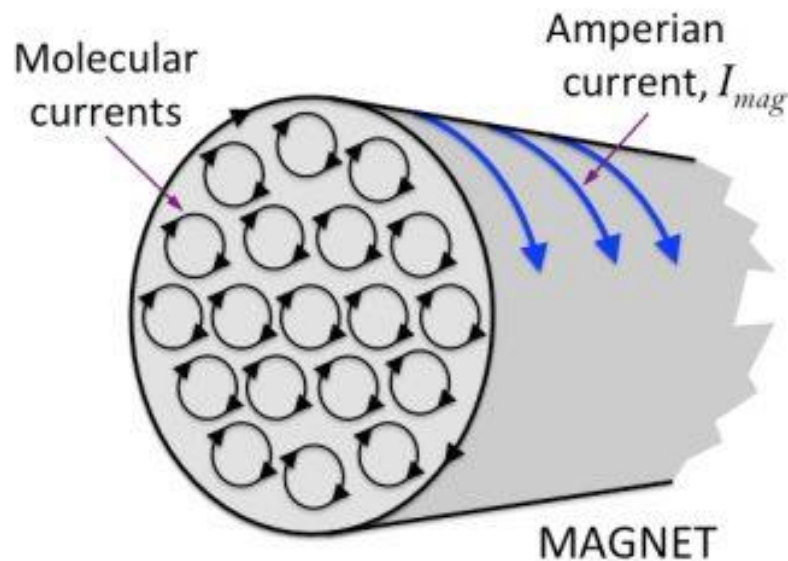
$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$



How about magnetic materials?

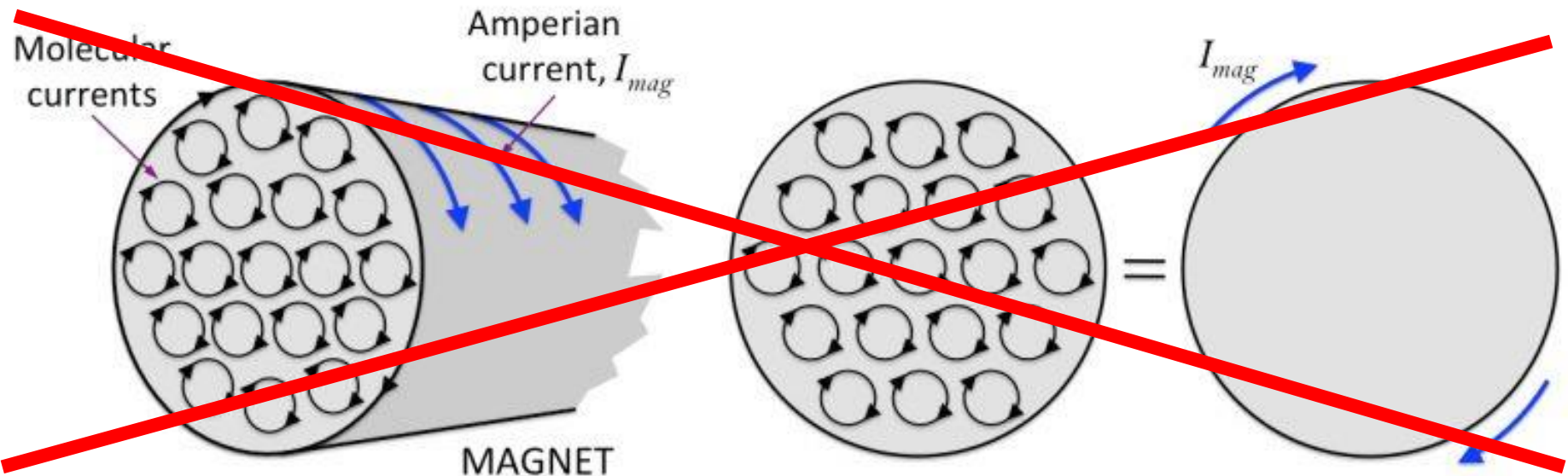
Origin of Magnetism - Old Theory

- Ampere 安培, 1826
 - Molecular Currents 分子电流假说
 - "magnetism is electricity in motion"



Origin of Magnetism - Old Theory

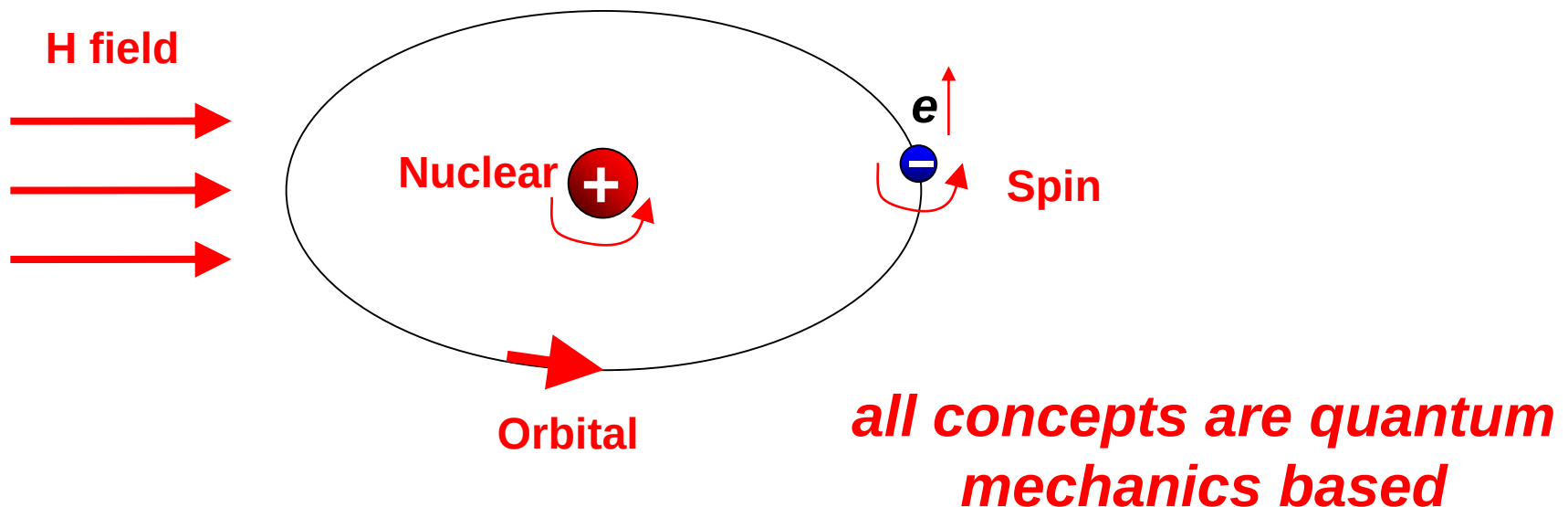
- However,
 - ❑ There are no “molecular currents” at all
 - ❑ For a steady solid, all the magnetic moments cancel out



***We can only understand magnetism
with quantum mechanics***

Origin of Magnetism - Modern Theory

- Magnetic moment of atoms
 - spin of electrons
 - orbital angular momentum
 - external magnetic field
 - magnetic momentum of nuclei (10^{-3} times smaller than that from electrons)



Electrodynamics

■ Maxwell's Equations

$$\nabla \cdot \mathbf{D} = \rho_V$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

Constitutive Relations

本构关系

$$\mathbf{B} = \mu_0 \mu_r \mathbf{H}$$

$$\mathbf{D} = \varepsilon_0 \varepsilon_r \mathbf{E}$$

$\varepsilon_0 \varepsilon_r$ - Permittivity (dielectric constant)

$\varepsilon_r = 1$ for vacuum

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

$\mu_0 \mu_r$ - Permeability

$\mu_r = 1$ for vacuum

$\mu_0 = 4\pi \times 10^{-7}$ H/m

Electrodynamics

■ Maxwell's Equations

$$\begin{aligned}\nabla \cdot \mathbf{D} &= \rho_V \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{H} &= \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}\end{aligned}$$

Constitutive Relations 本构关系

$$\begin{aligned}\mathbf{B} &= \mu_0 \mu_r \mathbf{H} \\ \mathbf{D} &= \varepsilon_0 \varepsilon_r \mathbf{E}\end{aligned}$$

For magnetic materials

$$\mu_r \neq 1$$

Magnetic Properties

- For magnetic materials, $\mu_r \neq 1$

$$\mu_r = \chi + 1$$

$$\mathbf{B} = \mu_0 \mu_r \mathbf{H} = \mu_0 (1 + \chi) \mathbf{H} = \mu_0 (\mathbf{H} + \mathbf{M})$$

$$\chi = \mu_r - 1$$

$$\mathbf{M} = \chi \mathbf{H}$$

- χ - Magnetic Susceptibility 磁化率
- B - Magnetic induction 磁感应强度
- H - Magnetic field 磁场强度
- M - Magnetization 磁化强度

Types of Magnetism

- χ - Magnetic Susceptibility 磁化率

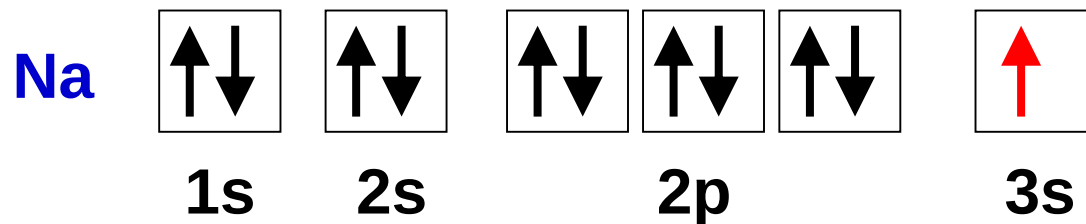
$$\mathbf{M} = \chi \mathbf{H}$$

- Diamagnetism 抗磁性/逆磁性 $\chi < 0$ $\sim 10^{-6}$
- Paramagnetism 顺磁性 $\chi > 0$ $10^{-4} \sim 10^{-5}$
- Ferromagnetism 铁磁性 $\chi \gg 0$ $> 10^{-2}$

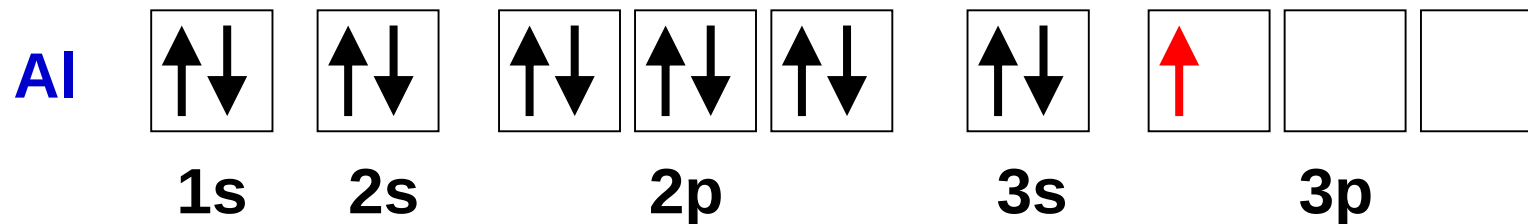
Paramagnetism 顺磁性

- Originated from *unpaired electrons*

▣ Sodium (Na) $[1s^2 2s^2 2p^6] 3s^1$

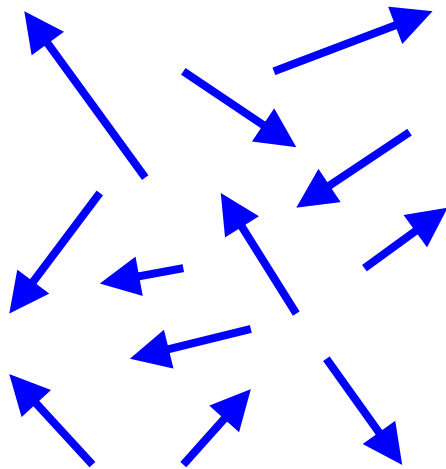


▣ Aluminum (Al) $[1s^2 2s^2 2p^6] 3s^2 3p^1$

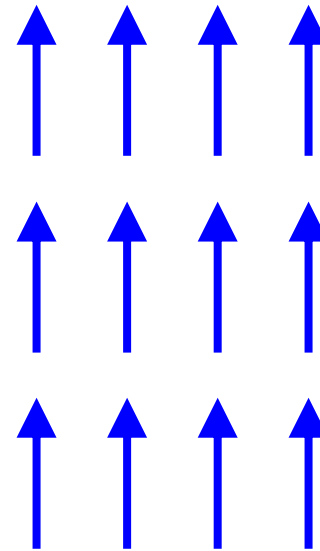


Paramagnetism 顺磁性

- Originated from *unpaired electrons*



no H field



with H field



H field

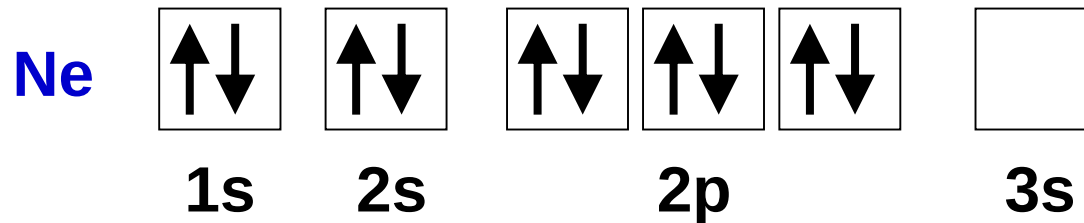
$$\mathbf{M} = \chi \mathbf{H}$$

Diamagnetism 抗磁性

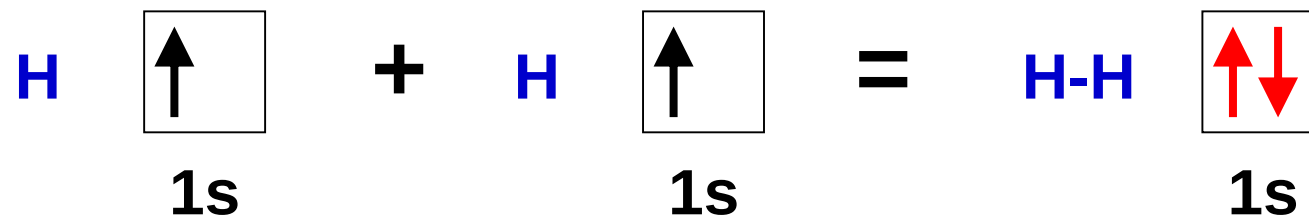
- Originated from *paired electrons*

▣ He, Ne, Ar, ...

Detailed analysis requires quantum mechanics



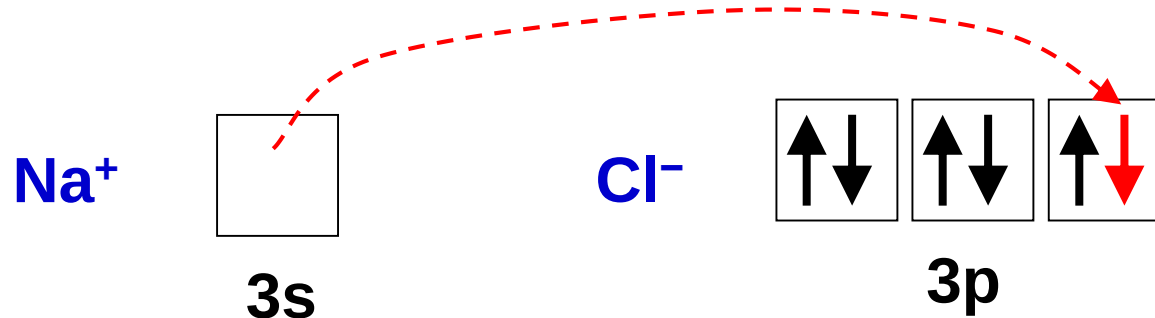
▣ H₂, N₂, ...



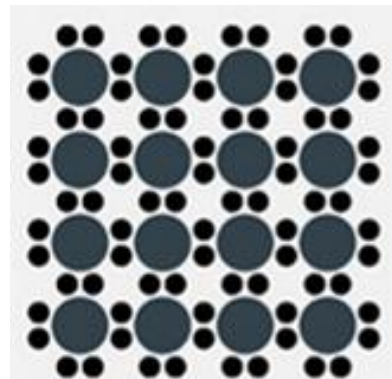
Diamagnetism 抗磁性

- Originated from *paired electrons*

□ $\text{NaCl} = \text{Na}^+\text{Cl}^-$

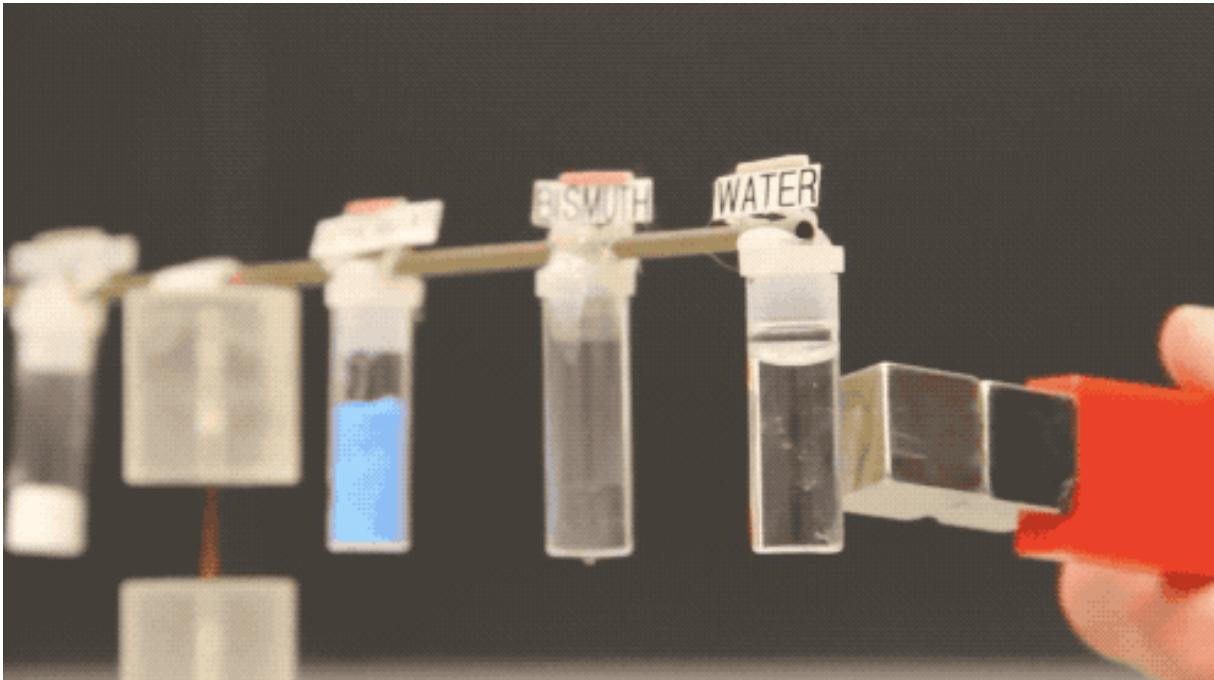
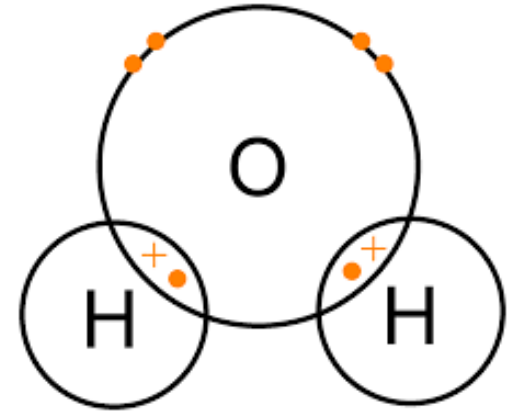


□ Silicon crystal



Diamagnetism 抗磁性

- Water (H_2O) is diamagnetic
 - ▣ All electrons are paired



Diamagnetism 抗磁性

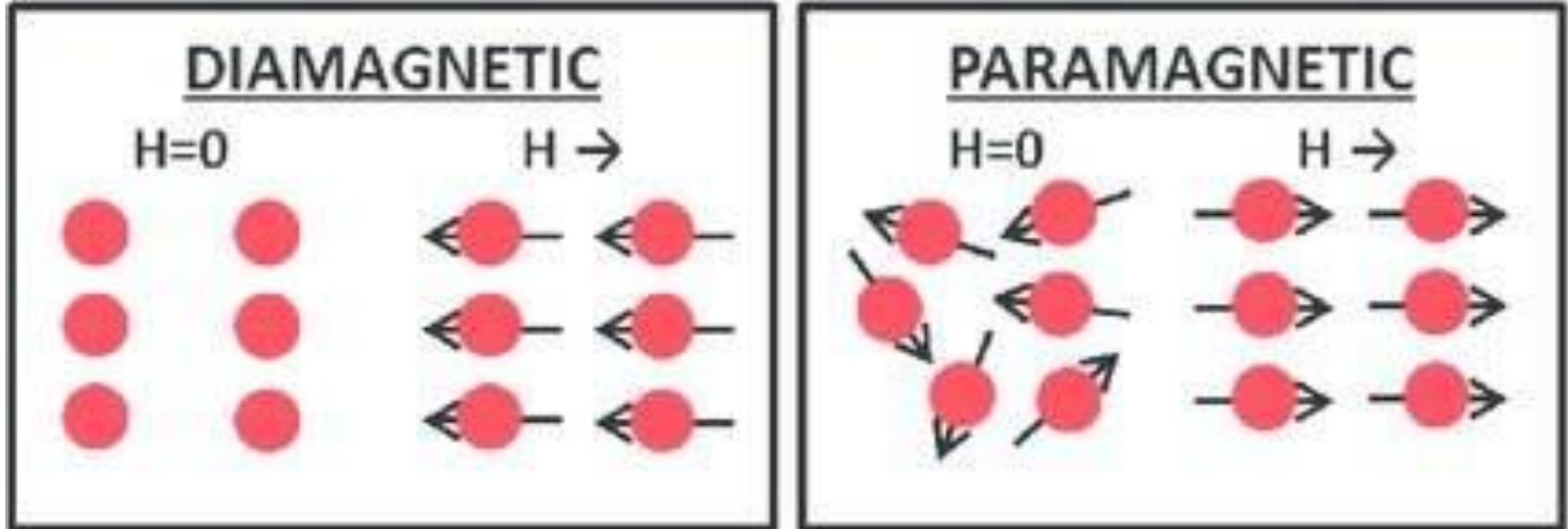
- Water (H_2O) is diamagnetic
 - A frog is lifted by a strong magnetic field ($H = 10 \text{ T}$)



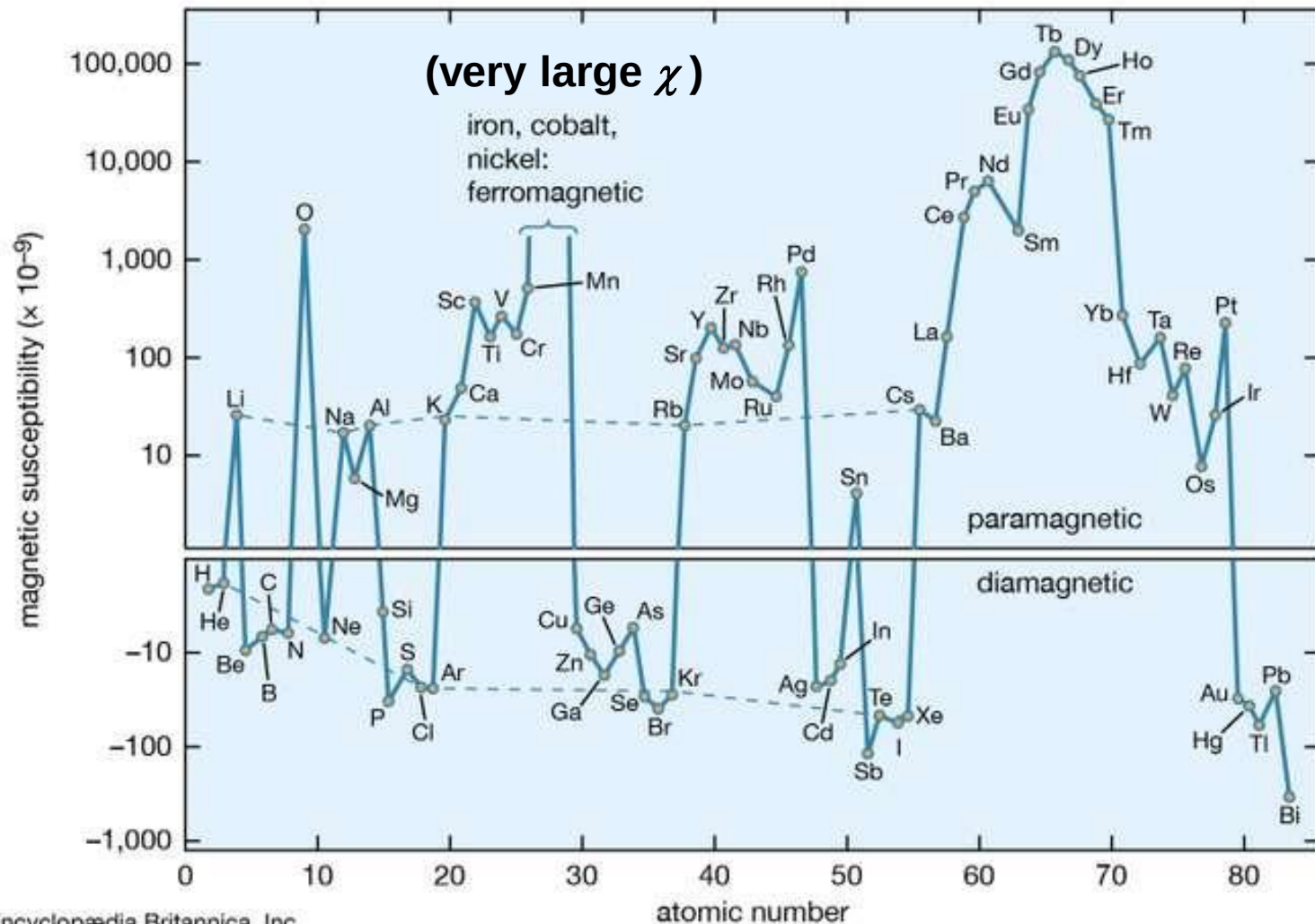
A. Geim
Nobel Prize in 2010
Ig Nobel Prize in 2001
(搞笑诺贝尔奖)

A. Geim, *Phys. Today* **51**, 9, 36 (1998)

Diamagnetism vs. Paramagnetism

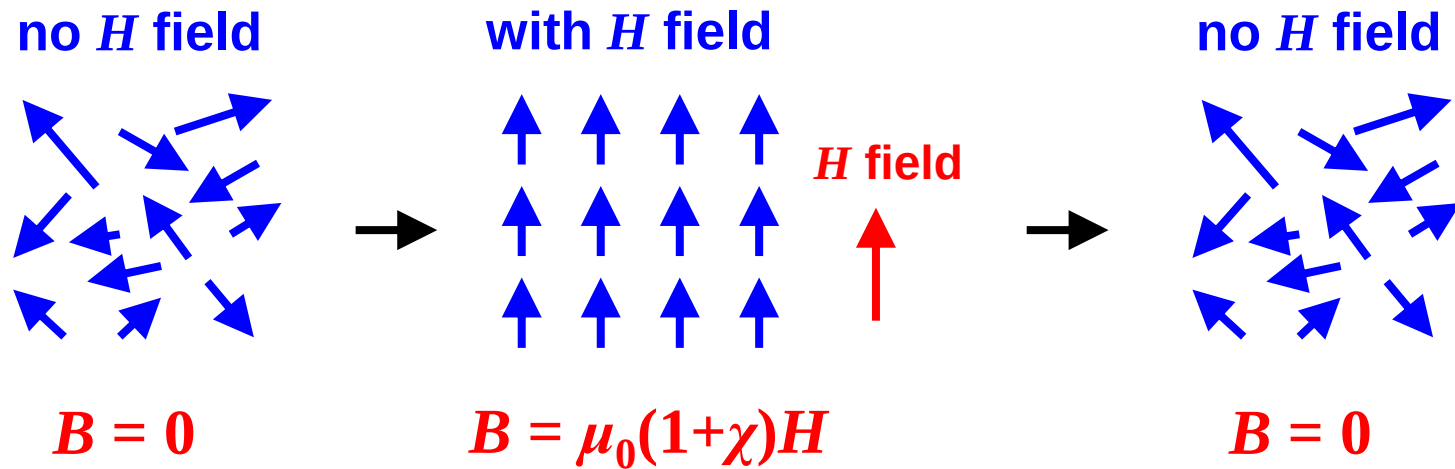


Diamagnetism vs. Paramagnetism



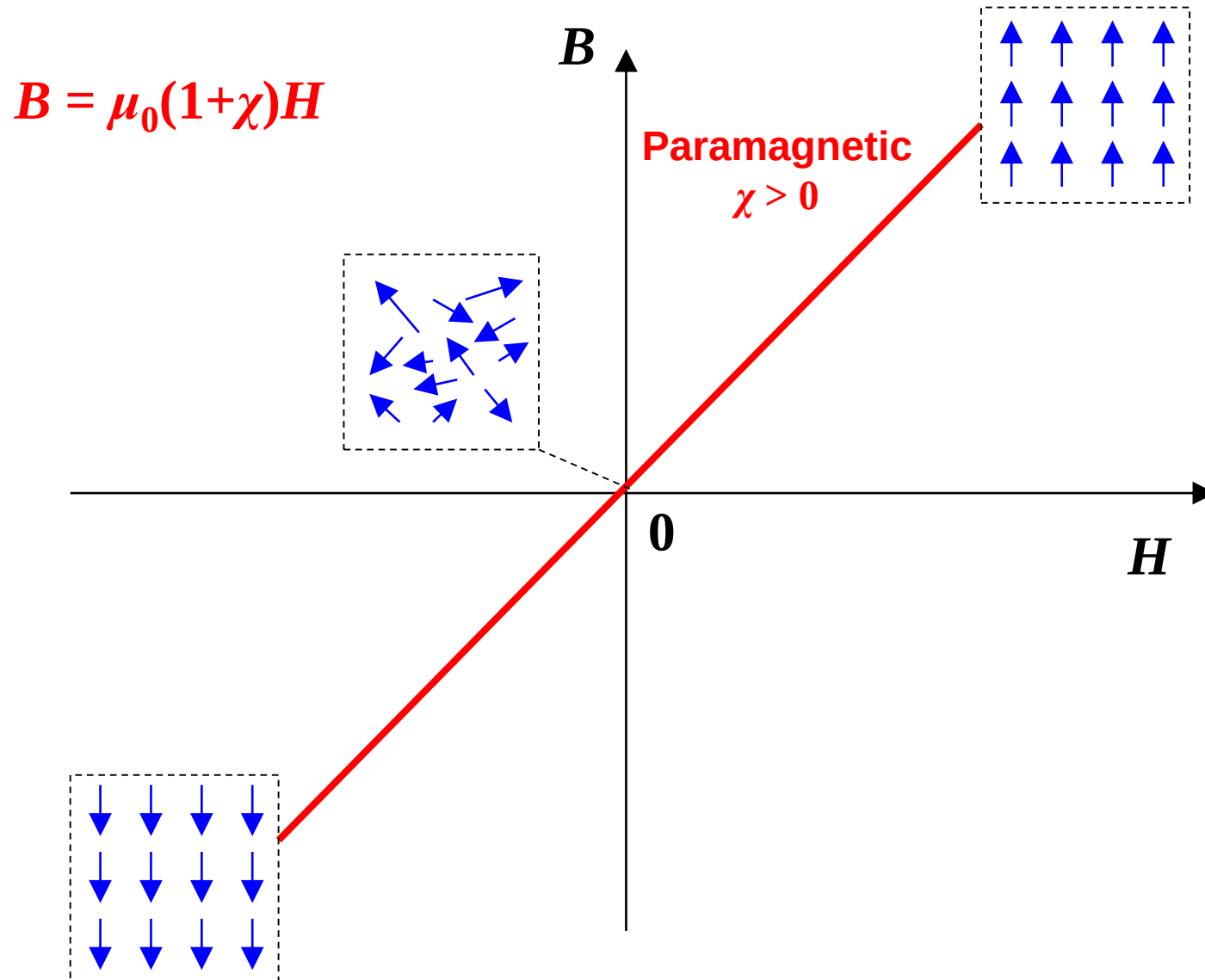
Magnetization Curve 磁化曲线

■ B vs. H

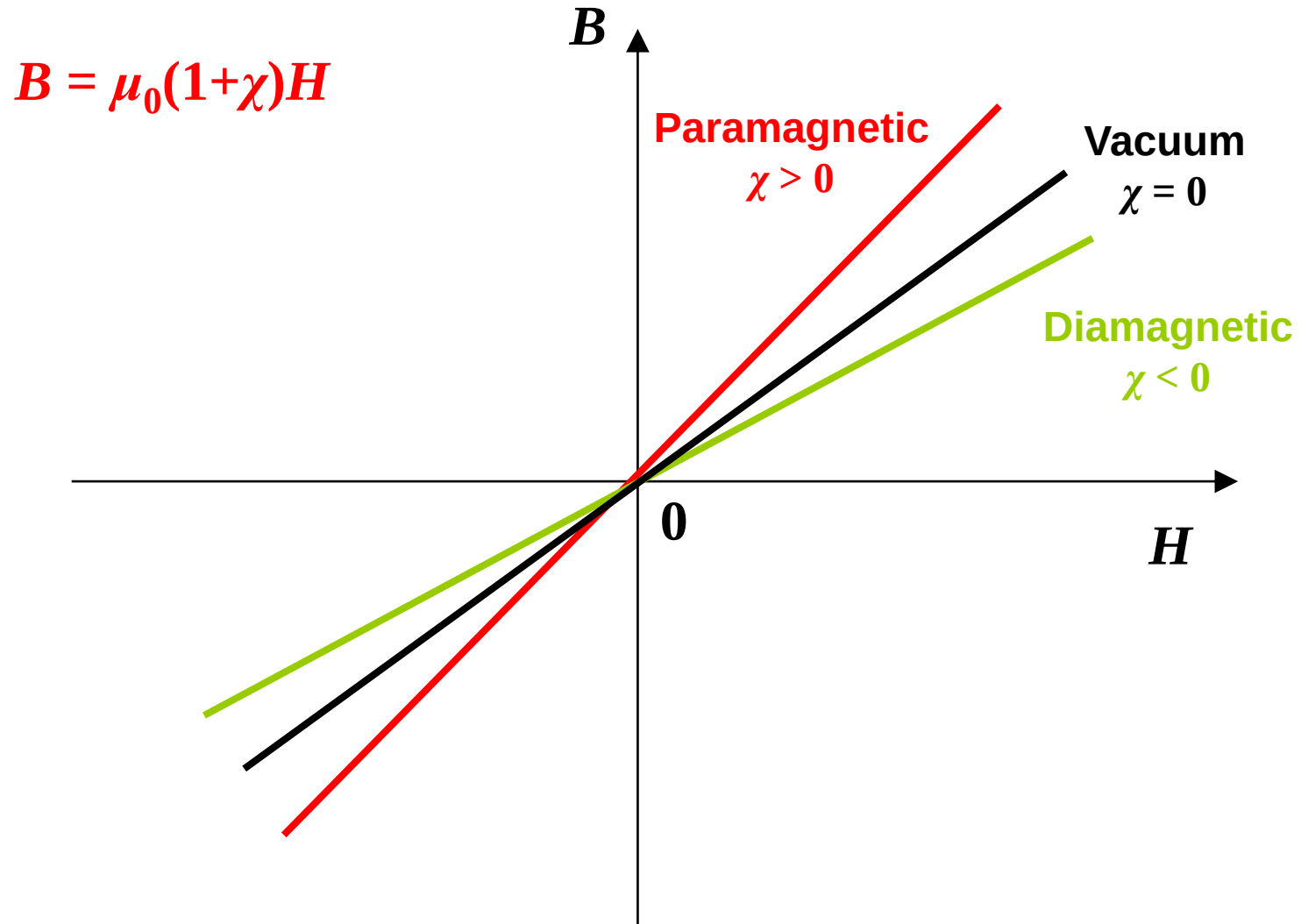


Paramagnetism

Magnetization Curve 磁化曲线



Magnetization Curve 磁化曲线



Ferromagnetism 铁磁性

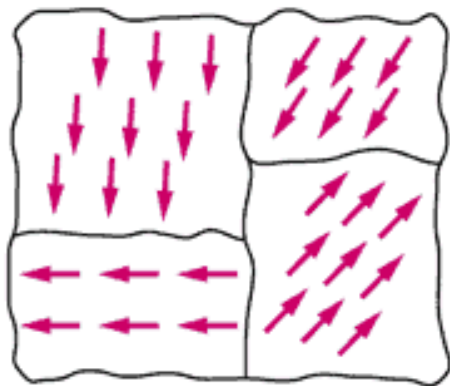
- Permanent Magnet (永磁体)
 - ▣ Fe, Co, Ni (铁, 钴, 镍)
 - ▣ Alloys: NdFeB (钕铁硼), SmCo (钐钴)



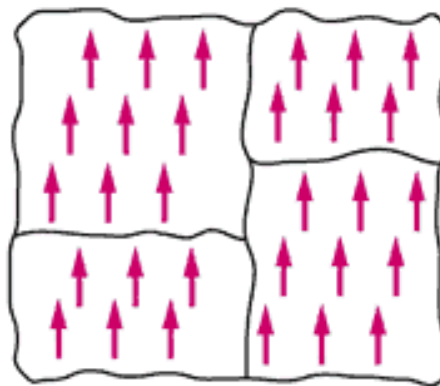
- Ferromagnetism 铁磁性 $\chi \gg 0$ $>10^{-2}$

Ferromagnetism 铁磁性

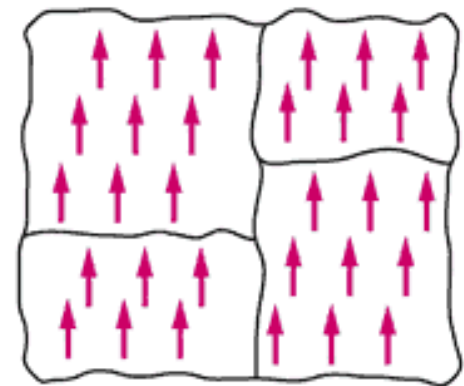
- When $H = 0$, magnetic domains (磁畴) form with spontaneous magnetization (自发磁化)
- Magnetization remains when H field is removed



no H field



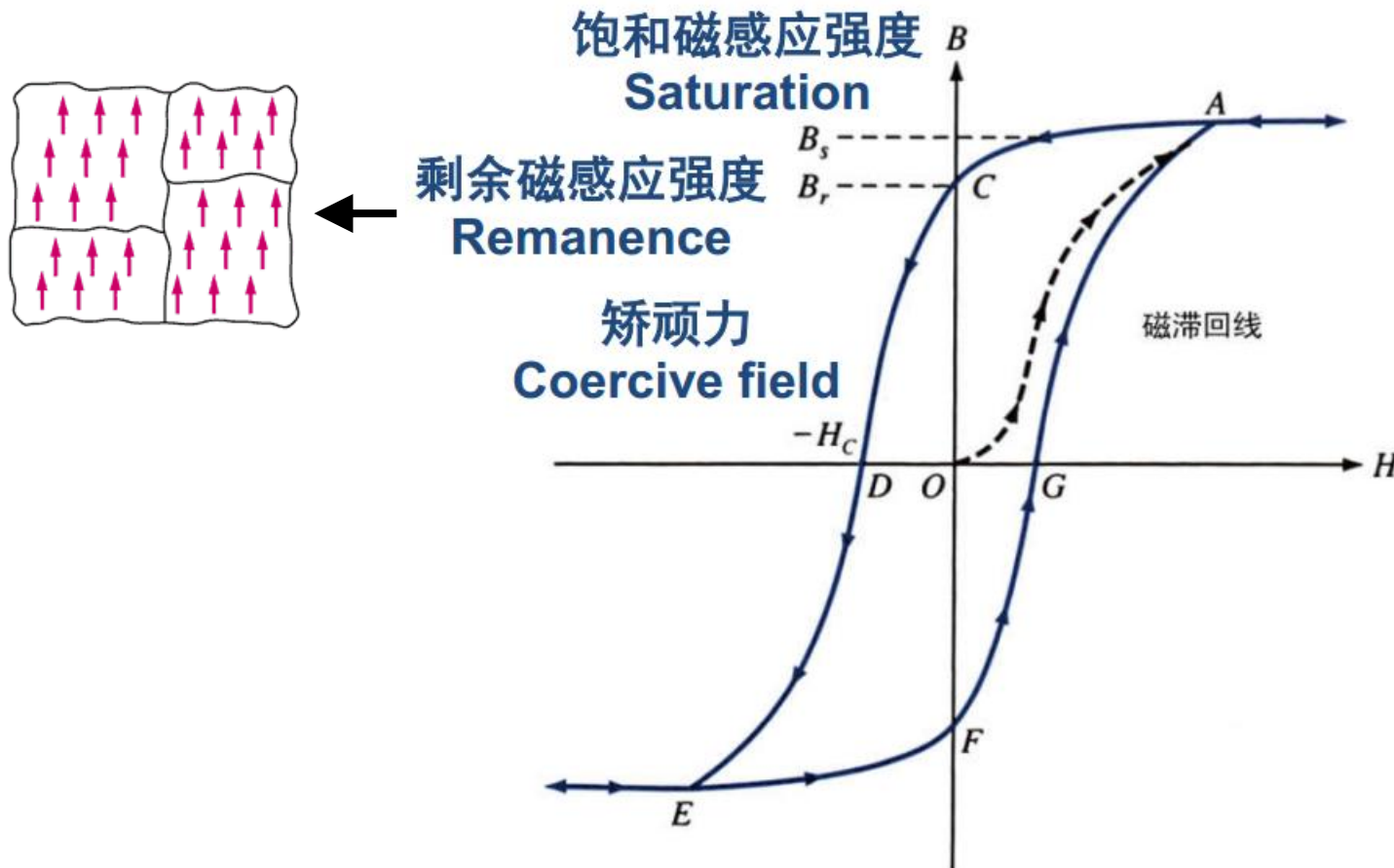
with H field



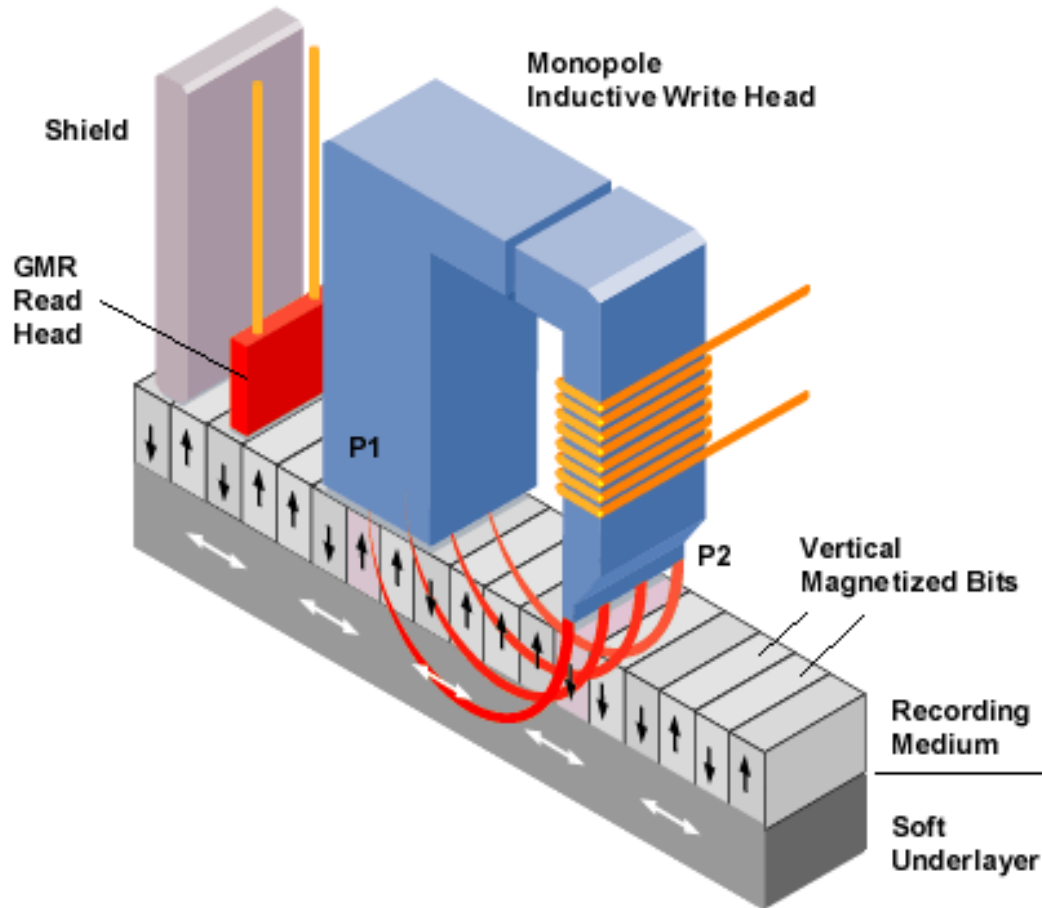
no H field

Ferromagnetism 铁磁性

- B - H curve forms a hysteresis loop (磁滞回线)



Magnetic recording



<https://encyclopedia2.thefreedictionary.com/Perpendicular+magnetic+recording>



Magnetic Tape 磁带



Hard Drive 硬盘

Evolution of Data Storage

Hard Drive 硬盘



5 MB
IBM 1956

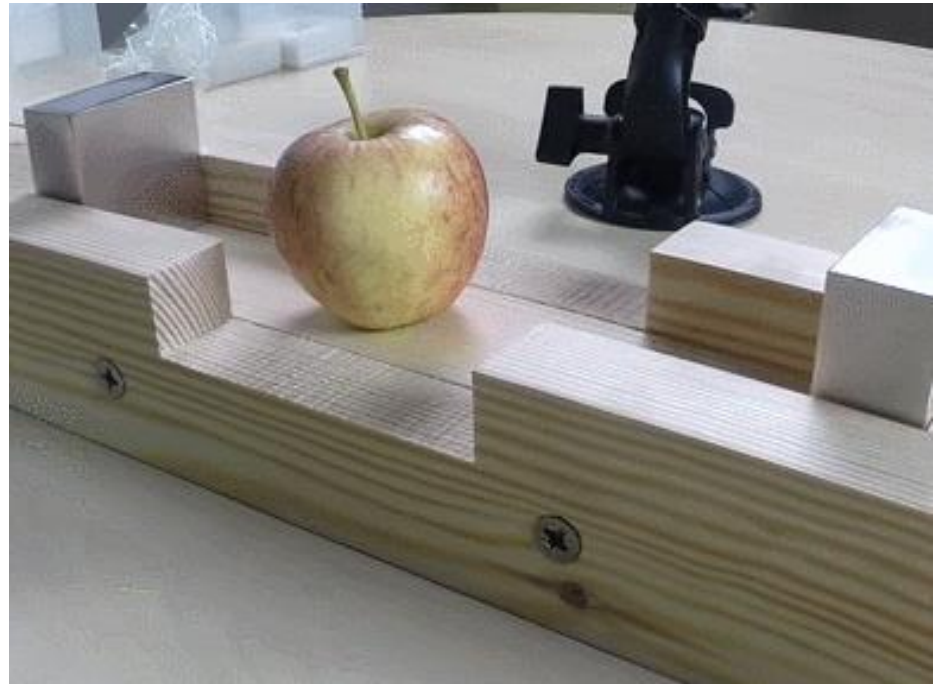


> 1 TB
Today

Ferromagnets can be Powerful



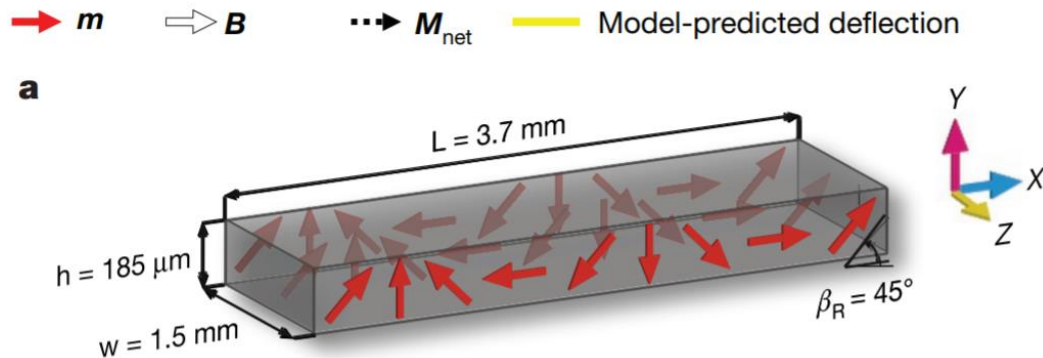
Be cautious!



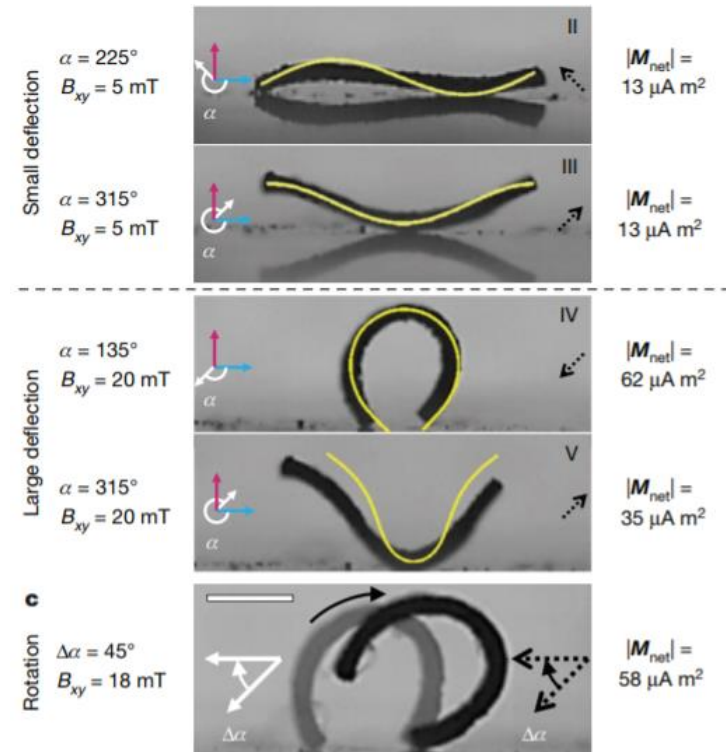
Mini Magnetic Robot

Small-scale soft-bodied robot with multimodal locomotion

Wenqi Hu^{1*}, Guo Zhan Lum^{1*}, Massimo Mastrangeli¹ & Metin Sitti¹

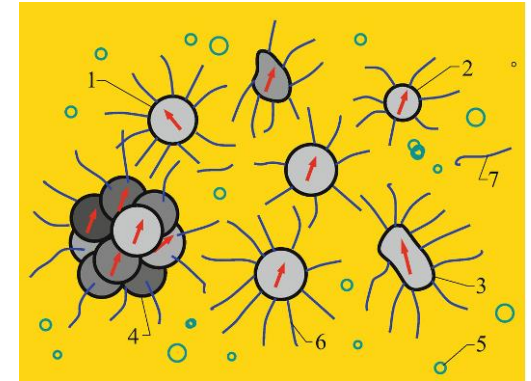


[Video](#)



Ferrofluid 铁磁流体

- A liquid with ferromagnetic particles



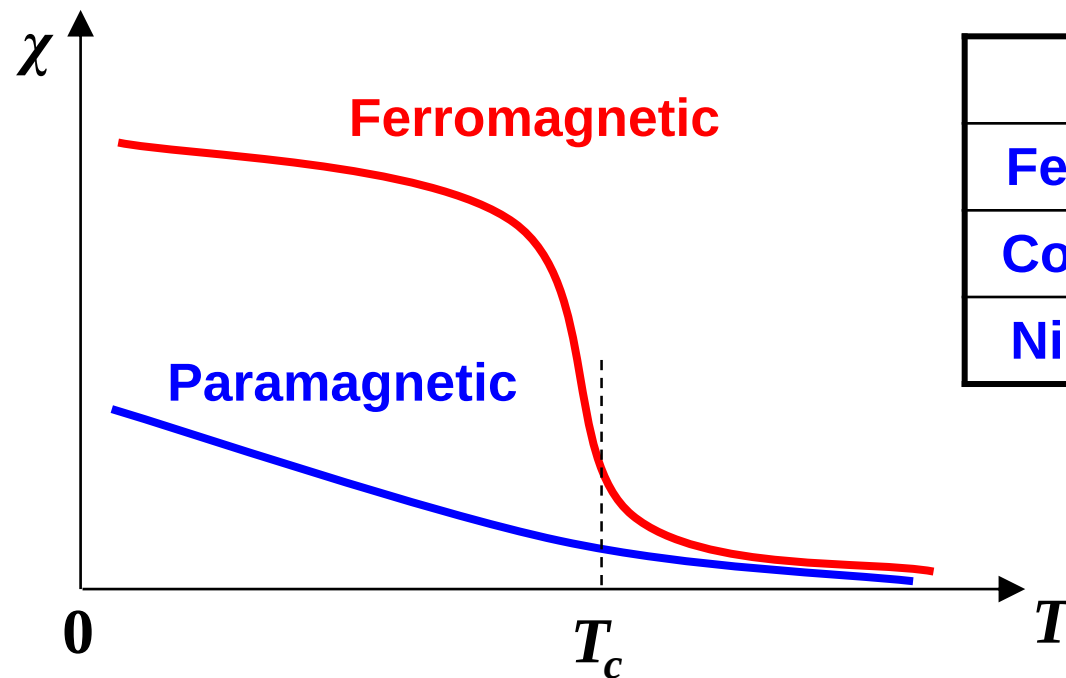
https://link.springer.com/chapter/10.1007/978-3-319-94427-2_1



<https://wonderfulengineering.com/these-15-magnet-gifs-will-show-you-the-power-of-magnetism/>

Temperature Effect

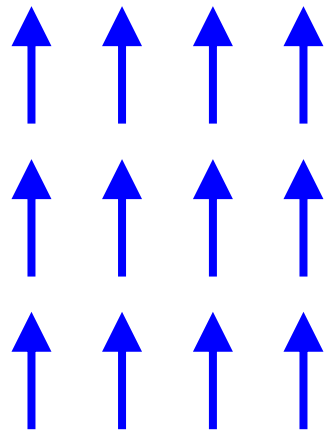
- Magnetization decreases with temperature, because of increased thermal fluctuation
- Ferromagnet becomes paramagnet when $T > T_c$
- T_c - Curie Temperature 居里温度



	T_c (°C)
Fe	770
Co	1110
Ni	350

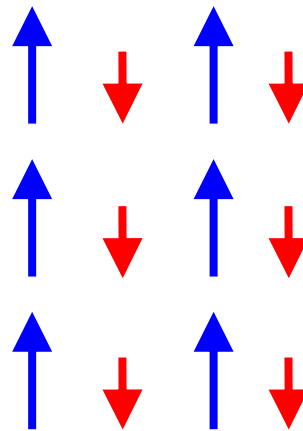
Antiferromagnetism and Ferrimagnetism

- **Ferri**magnetism 亚铁磁性
- **Antiferro**magnetism 反铁磁性



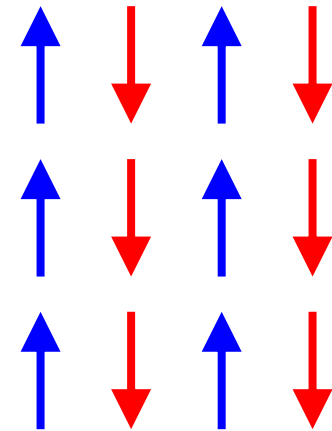
Ferromagnetic

$$\chi_1 \gg 0$$



Ferrimagnetic

$$\chi_2 < \chi_1$$

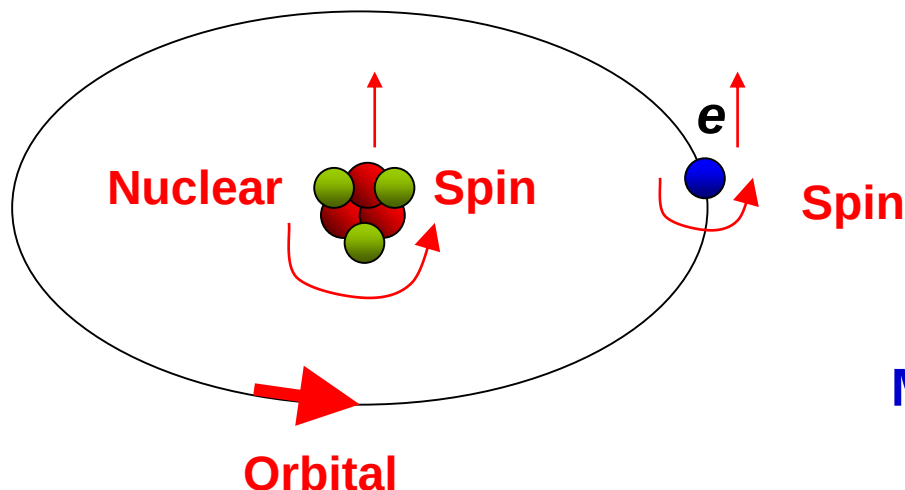


Antiferromagnetic

$$\chi_3 \ll \chi_1$$

Magnetic Properties of Nuclei

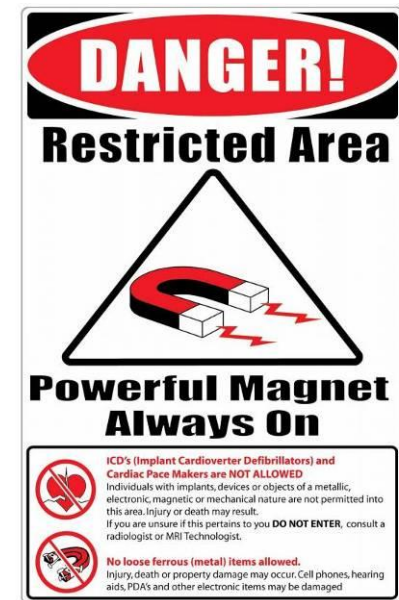
- Protons (质子) and Neutrons (中子) in the nuclei also have spins that generate magnetic moments
- Nuclei with odd numbers of protons and neutrons have a net magnetic moment
- χ is much smaller ($< 1/1000$) than those of electrons



Magnetic Resonance Imaging (MRI)
核磁共振成像
detect ^1H atoms

Be cautious when doing MRI

- Very strong magnetic field



Summary

- **Maxwell's Equations**

- H, B, M, μ_r
- **Magnetic Susceptibility 磁化率 χ**

- **Origin of magnetism**

- spin of electrons, orbital angular momentum, external field
- nuclear magnetic momentum

- **Types of magnetism**

- **Diamagnetism 抗磁性**
- **Paramagnetism 顺磁性**
- **Ferromagnetism 铁磁性**
- **Antiferromagnetism 反铁磁性**
- **Ferri magnetism 亚铁磁性**

Thank you for your attention